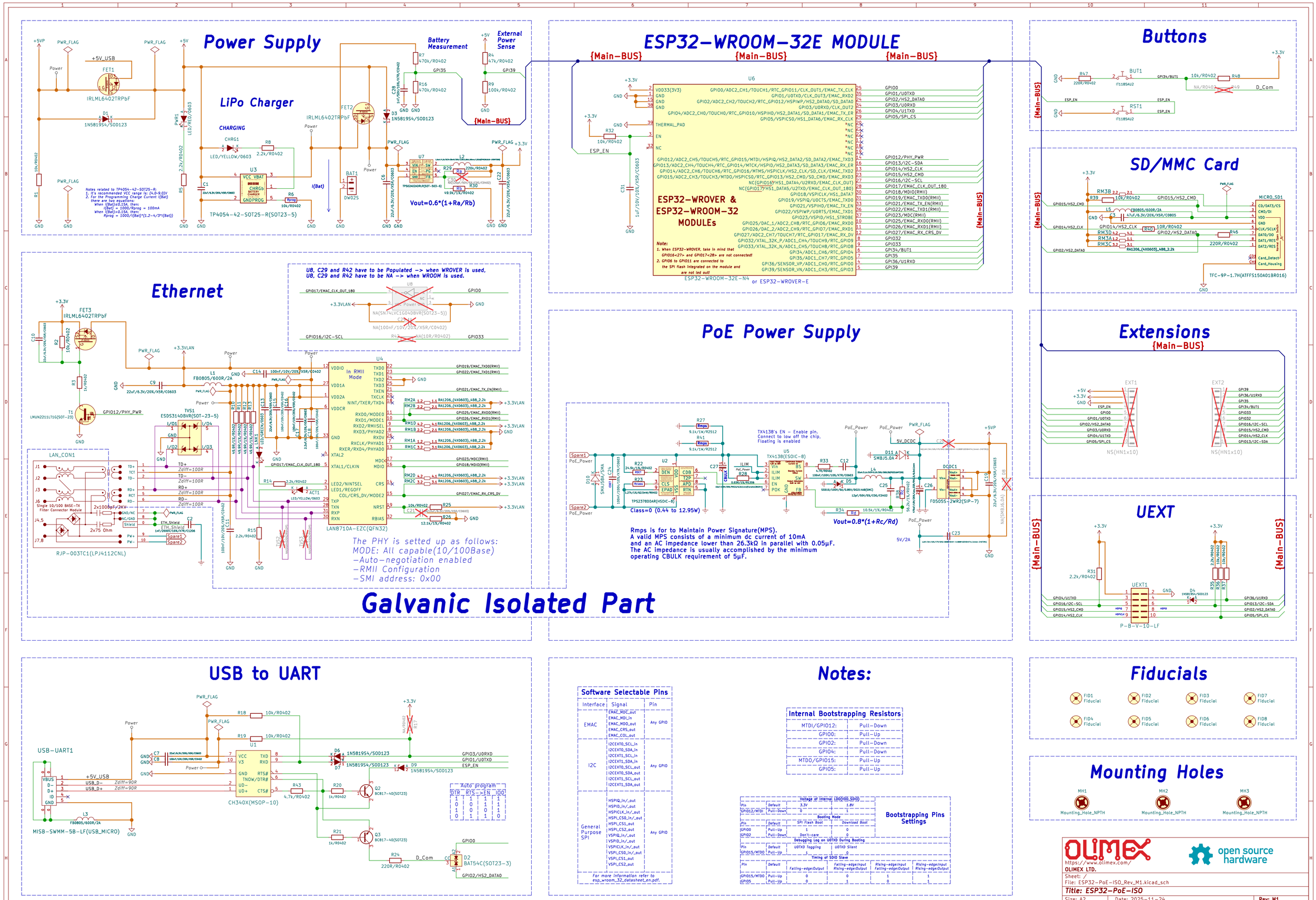
Power Supply
LiPo Charger
CHARGING
Battery Measurement
External Power Sense
Vout=0.6*(1+Ra/Rb)
TP4054-42-SOT25-R(SOT23-5)
ESP32-WROOM-32E MODULE
{Main-BUS}
ESP32-WROVER & ESP32-WROOM-32 MODULEs
ESP32-WROOM-32E-N4 or ESP32-WROVER-E
Buttons
SD/MMC Card
MICRO_SD1
TFC-9P-1.7H(ATFFS150A01BR016)
Ethernet
U8, C29 and R42 have to be Populated -> when WROVER is used,
U8, C29 and R42 have to be NA -> when WROOM is used.
LAN8710A-EZC(QFN32)
RJP-003TC1(LPJ4112CNL)
The PHY is setted up as follows:
MODE: All capable(10/100Base)
-Auto-negotiation enabled
-RMII Configuration
-SMI address: 0x00
PoE Power Supply
TX4138's EN - Enable pin.
Connect to low off the chip.
Floating is enabled
Class=0 (0.44 to 12.95W)
Vout=0.8*(1+Rc/Rd)
Rmps is for to Maintain Power Signature(MPS).
A valid MPS consists of a minimum dc current of 10mA
and an AC impedance lower than 26.3kΩ in parallel with 0.05µF.
The AC impedance is usually accomplished by the minimum
operating CBULK requirement of 5µF.
Extensions
UEXT
Galvanic Isolated Part
USB to UART
USB-UART1
MISB-SWMM-5B-LF(USB_MICRO)
CH340X(MSOP-10)
Auto program
Notes:
Software Selectable Pins
Internal Bootstrapping Resistors
Bootstrapping Pins Settings
Fiducials
Mounting Holes
OLIMEX LTD.
https://www.olimex.com/
open source hardware
Sheet: /
File: ESP32-PoE-ISO_Rev_M1.kicad_sch
Title: ESP32-PoE-ISO
Size: A2
Date: 2025-11-24
Rev: M1
KiCad E.D.A. kicad 7.0.9-7.0.9-ubuntu22.04.1
Id: 1/1